

CABANATUAN, Philippines

WMO: 983300

Lat: 15.471N

Lon: 120.952E

Elev: 32

StdP: 100.94

Time Zone: 8.00 (E08)

Period: 96-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	21.0	17.5	12.6	23.3	18.3	13.3	23.4	6.4	28.0	5.6	27.8	0.4	0	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	9.4	36.3	28.8	35.5	28.6	34.6	28.5	30.2	34.4	29.8	33.8	29.3	33.2	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	26.0	33.5	28.7	25.3	33.0	28.3	24.7	32.4	101.2	34.6	99.0	33.9	96.6	33.3	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.4	4.4	3.6	17.7	37.8	2.9	0.6	15.6	38.2	13.9	38.6	12.3	38.9	10.2	39.4		
			16.7	31.4	1.8	1.0	15.4	32.2	14.4	32.8	13.4	33.4	12.0	34.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	26.2	26.9	28.0	29.7	29.9	28.9	28.2	27.9	28.0	28.0	27.6	27.0
	DBStd	1.65	1.30	1.26	1.26	1.35	1.48	1.29	1.30	1.23	1.18	1.17	1.20	1.37
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6581	503	472	559	590	617	568	564	553	539	558	529	527
	CDD18.3	3539	245	239	301	340	359	318	306	295	289	299	279	269
	CDH23.3	36208	2012	2143	3119	4138	4304	3528	3107	2874	2842	2989	2709	2443
	CDH26.7	14320	678	823	1372	2084	2096	1457	1091	896	946	1072	987	818
(14) Wind	WSAvg	1.0	1.2	1.2	1.1	0.9	1.0	0.9	0.9	0.9	0.8	1.1	1.0	1.4
(15) Precipitation	PrecAvg	1920	6	8	18	39	169	251	317	388	298	223	105	65
	PrecMax	2651	24	49	123	236	587	591	608	691	629	668	344	523
	PrecMin	1306	0	0	0	0	14	90	94	199	120	8	1	0
	PrecStd	363	6	13	26	52	117	131	123	137	109	152	92	97
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	33.9	35.4	37.1	37.6	35.8	34.5	33.9	33.9	34.1	33.8	33.1
		MCWB	27.2	27.5	28.3	28.9	29.1	29.2	29.7	29.5	29.0	28.5	27.9	27.4
	2%	DB	31.7	32.9	34.3	36.2	36.6	34.7	33.6	33.0	33.0	33.2	33.0	32.2
		MCWB	26.6	27.3	28.0	28.7	29.0	29.1	29.1	28.9	28.6	28.1	27.6	27.1
	5%	DB	31.0	32.0	33.5	35.5	35.6	33.8	32.7	32.0	32.1	32.4	32.2	31.5
		MCWB	26.3	26.9	27.7	28.4	28.7	28.9	28.7	28.4	28.2	27.8	27.3	26.7
	10%	DB	30.2	31.1	32.5	34.6	34.6	32.8	31.7	31.0	31.2	31.5	31.4	30.7
		MCWB	25.7	26.3	27.2	28.2	28.5	28.6	28.2	27.9	27.8	27.5	27.0	26.2
	0.4%	WB	28.2	29.1	29.7	30.5	30.9	30.6	30.3	30.1	30.1	29.9	29.3	29.0
		MCDB	31.5	32.5	33.8	35.6	36.1	34.7	33.7	33.1	32.9	32.9	32.5	32.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.5	28.1	28.9	29.8	30.1	29.8	29.7	29.3	29.2	29.1	28.6	28.1
		MCDB	30.9	31.9	33.3	35.0	35.1	33.8	33.0	32.2	32.1	32.1	31.9	31.4
	5%	WB	26.8	27.4	28.2	29.1	29.5	29.2	29.0	28.7	28.7	28.5	28.0	27.3
		MCDB	30.3	31.3	32.6	34.3	34.4	33.0	32.2	31.5	31.5	31.5	31.4	30.7
	10%	WB	26.1	26.6	27.5	28.4	28.8	28.7	28.3	28.1	28.1	27.9	27.4	26.7
		MCDB	29.7	30.5	31.9	33.7	33.6	32.3	31.2	30.6	30.7	30.8	30.7	29.9
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	9.6	10.2	9.4	7.8	6.8	6.1	6.7	7.3	7.8	7.6
		MCDBR	9.4	10.2	11.0	11.4	11.1	9.4	8.6	7.9	8.3	8.7	9.2	8.7
		MCWBR	5.4	5.6	5.7	5.3	4.8	4.8	4.9	4.6	4.6	4.6	4.8	4.7
	5% WB	MCDBR	8.8	9.6	10.2	10.8	10.3	8.8	8.2	7.5	7.8	8.1	8.6	8.3
		MCWBR	5.6	6.1	6.0	5.8	5.5	5.0	5.0	4.7	4.9	5.0	5.2	5.2
(40) Clear-Sky Solar Irradiance	taub		0.410	0.415	0.454	0.493	0.485	0.467	0.463	0.473	0.470	0.474	0.431	0.428
	taud		2.418	2.391	2.281	2.183	2.234	2.298	2.314	2.278	2.282	2.256	2.396	2.386
	Ebn at Noon		881	894	864	827	821	827	832	832	838	826	854	852
	Edh at Noon		113	121	138	152	143	132	131	137	136	136	115	114
(44) All-Sky Solar Radiation	RadAvg		4.29	4.99	5.58	6.17	5.72	5.20	4.65	4.33	4.74	4.57	4.21	3.92
	RadStd		0.21	0.39	0.33	0.31	0.48	0.55	0.56	0.44	0.42	0.39	0.41	0.29

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)	+0.32	+0.57	N/A	+0.40	+0.36	+0.42	N/A	N/A	+158	+158

Nomenclature: See separate page